

Lydia Giacomini

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Education

- **Postdoc in Random Matrix Theory at King's College London** 2025-
with Yan Fyodorov
ESPRC-UKRI grant
- **PhD** 2022-2025
'A renormalization group approach to transport phenomena'
Under the supervision of François Huveneers (Université Paris Dauphine)
- **Master's degree in Probability and Statistics** 2020-2022
Ecole Normale Supérieure de Paris and Université d'Orsay
- **First year of ICFP Master in Physics** 2019-2020
Ecole Normale Supérieure de Paris
- **Bachelor's degree in Physics** 2018-2019
Ecole Normale Supérieure de Paris
- **Classes Préparatoires Scientifiques** 2016-2018
Lycée Henri IV
- **General Education Degree (Baccalauréat scientifique)** June 2016

Research activity

- **A renormalization group approach to transport phenomena** 2025
PhD thesis
- **Absence of Normal Heat Conduction in Strongly Disordered Interacting Quantum Chains** 2024
<https://arxiv.org/pdf/2408.04338>
Pre-print, joint work with François Huveneers (Université Paris Dauphine), Wojciech de Roeck (KU Leuven) and Oskar Prosnia (KU Leuven)
- **Sub-Power Law Decay of the Wave Packet Maximum in Disordered Anharmonic Chains** 2025
<https://arxiv.org/pdf/2502.02344> (submitted to *Nonlinearity*, accepted with minor revisions as of Dec 2025)
Proof of upper bounds on the speed of transport of energy in a disordered chain of oscillators with a quartic interaction potential
Joint work with Wojciech De Roeck (KU Leuven), Amirali Hannani (KU Leuven) and François Huveneers (Université Paris Dauphine)

- **Aspects of the Euclidean Log-Gas (ongoing)**

A study of the properties of the Euclidean log-gas, based on prior work by John E. McCarthy (<https://arxiv.org/pdf/2305.20029>) establishing a link between d-tuples of GUE matrices and the Euclidean d-dimensional log-gas

Joint work with Djalil Chafaï (Université Paris Dauphine) and David García-Zelada (Sorbonne Université)

- **Spectral properties of non-Hermitian Random Matrices (ongoing)**

This ongoing project aims to characterize the spectral behavior of a few non Hermitian random matrix models (joint distribution of left and right complex GinUE eigenvectors ; eigenvalue and eigenvector characterization of perturbed GUE matrices) using supersymmetry techniques.

Joint work with Yan Fyodorov (King's College London)

- **Collective modes for complex GinUE right eigenvectors (ongoing)**

In this ongoing project, I propose to study the collective behavior of eigenvectors of the complex GinUE ensembles (ie. high-dimensional matrices with iid. Gaussian entries). To this end, I investigate the empirical covariance matrix of the GinUE eigenvectors in order to uncover eigenvector-eigenvector correlations.

Internships and other academic experience

- **Stochastic Dynamics of Extended Quantum Systems, in and out of Equilibrium** *April - July 2022*

Based on the following paper by Denis Bernard (Ecole Normale Supérieure de Paris):
<https://arxiv.org/pdf/2107.04442>

A study of toy models for quantum simple exclusion processes (QSEPs), and quantum parallels for classical quantities that characterize Macroscopic Fluctuation Theory (MFT)

Under the supervision of Denis Bernard (Ecole Normale Supérieure de Paris)

- **Stochastic Growth Models With Connections to Random Matrix Theory** *April - July 2020*

A study of the parallels between the TASEP and the eigenvalue distribution of GOE/GUE matrices. Scaling limit of the distribution of the largest GUE eigenvalue, convergence in law to a Tracy-Widom distribution, and parallels with the position of particles in the TASEP in the long-time limit

Under the supervision of Patrik L. Ferrari (Universität Bonn)

Talks given

- **Young Researchers' Days seminar** *June 2023*

Organized by Ceremade, Université Paris Dauphine

- **Matinée des doctorants (PhD students' day)** *April 2025*

Third year PhD students' talks organized by Ceremade, Université Paris Dauphine

- **Bristol Mathematical Physics Seminar** *Dec 2025*

Localization in certain disordered quantum spin chains

- **Young Researchers' Seminar** *Dec 2025*

Localization in certain disordered quantum spin chains

Ceremade, Université Paris Dauphine

Teaching experience

- **Teaching assistant, Poisson processes** (first-year Master's course) 2022-2025
Université Paris Dauphine
- **Teaching assistant, Probability** (second-year undergraduate course) 2022-2024
Université Paris Dauphine
- **Research workshop, Rendez-Vous des Jeunes Mathématiciennes** 14-15 Dec 2024
The RVJM are a weekend-long event for high school girls with the purpose of introducing them to the different fields of mathematics. As part of this event, I facilitated a "research workshop" where participants had to work in small groups on an open problem and then present their results to the other groups
Ecole Normale Supérieure de Paris

Skills

- **Languages:** French (fluent), English (fluent), Italian (fluent), Farsi (basic conversations), German (B1)
- **Programming:** Python, R, Matlab

Other

- **Music:** studied the piano at the Ecole Normale de Musique de Paris (2012-2016)
- **Sports:** distance running (national level in cross-country and track)